CSMSS

CHH.SHAHUCOLLEGE OF ENGINEERING

Kanchanwadi,PaithanRoad, Chhatrapati Sambajinagar

Maharashtra

Seminar Presentation

A presentation of seminar

ON

Programming of Power Factor control using Arduino

UNDER the Guidance of

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INTRODUCTION

Power factor is the ratio between real power and apparent power in an electrical system. In industries and electrical installations, low power factor causes higher current flow, increased power losses, voltage drop, and electricity penalties.

To improve the power factor, capacitor banks are used. Manual switching of capacitors is difficult and inefficient. Therefore, automatic power factor correction systems are used.

Arduino is a low-cost microcontroller platform that can monitor voltage and current continuously and automatically control capacitor switching for power factor improvement.

This project focuses on programming an Arduino-based automatic power factor control system.

Problem Identification

Industries and electrical systems face several problems due to low power factor:

- Increase in power losses
- Higher electricity bills
- Voltage fluctuations
- Overheating of equipment
- Reduced efficiency of electrical systems
- Manual control of capacitor banks is time-consuming

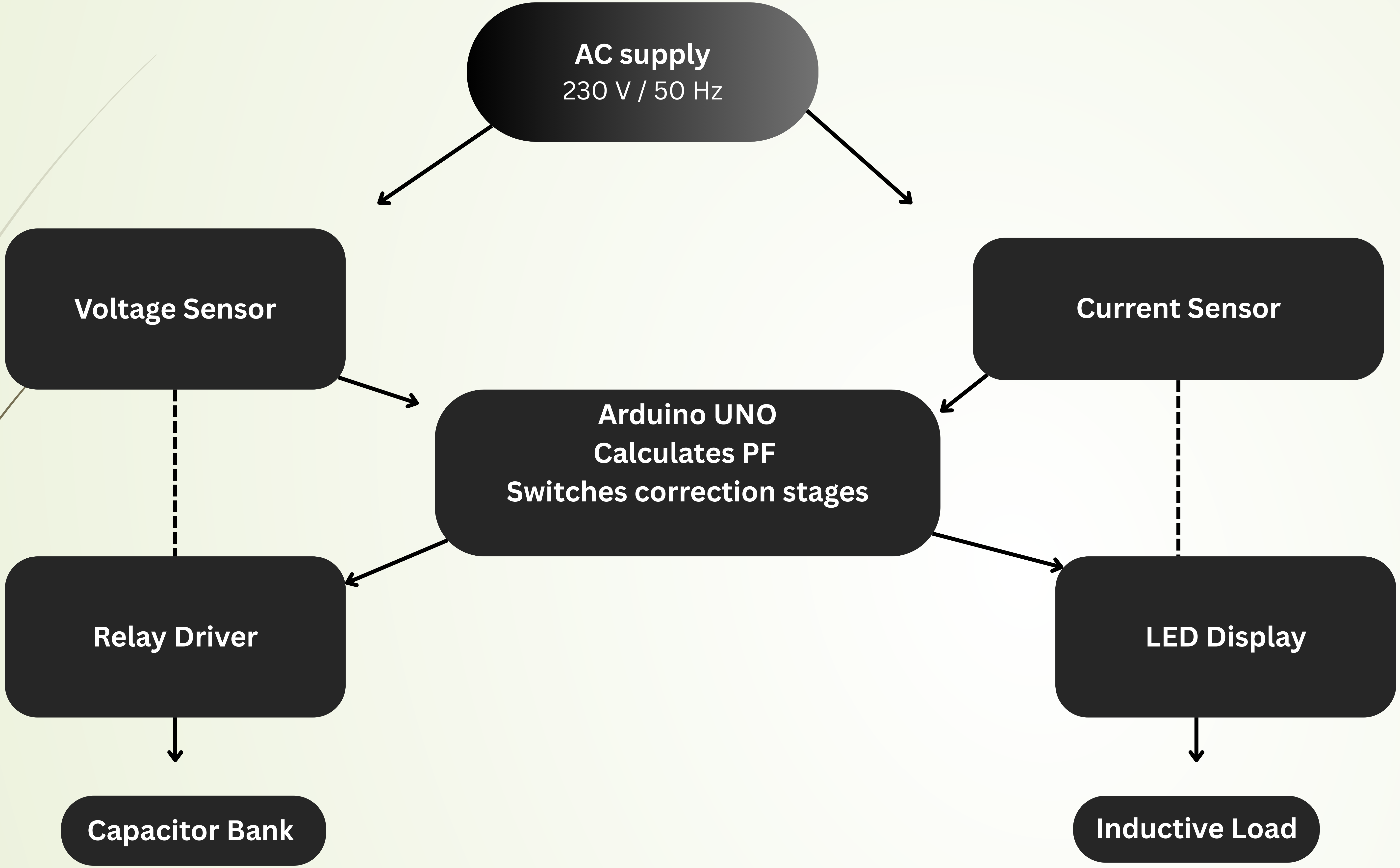
Existing systems are expensive and less flexible. Hence, there is a need for a low-cost automatic solution using Arduino.

OBJECTIVES

The main objectives of this project are:

- To measure power factor using Arduino
- To improve power factor automatically
- To reduce power losses and electricity cost
- To automate capacitor bank switching
- To design a low-cost and efficient system
- To understand Arduino programming in electrical applications

METHODOLOGY



ADVANTAGES

- Automatically improves power factor without manual operation
- Reduces electricity bill and power factor penalty charges
- Minimizes power losses in transmission lines
- Improves efficiency of electrical equipment
- Increases life of motors and electrical devices
- Fast and accurate switching of capacitor banks
- Low-cost system compared to industrial APFC panels
- Compact and easy to install
- Arduino programming can be modified easily as per requirement
- Provides real-time monitoring and control

DISADVANTAGES

- Accuracy depends on sensor quality and calibration
- Arduino has limited processing capability for large industrial systems
- Capacitor switching may create harmonics in some conditions
- Requires proper programming and circuit protection
- System may fail if sensors or relays malfunction
- Not suitable for very high voltage applications without additional protection circuits

APPLICATIONS

- Industrial power systems
- Manufacturing plants
- Commercial buildings
- Electrical laboratories
- Distribution systems
- Motor load applications
- Smart energy management systems

CONCLUSION

Arduino-based power factor control is an efficient and economical solution for improving electrical system performance. The system automatically detects low power factor and controls capacitor banks to maintain efficient operation. It reduces losses, improves efficiency, and minimizes electricity cost.

This project also demonstrates the practical application of embedded systems in electrical engineering.

Thank you !